

# **Sentiment Prediction of Text Review**

**A Project Report**

Submitted in partial fulfilment of the  
requirements for the award of the Degree of

**BACHELOR OF SCIENCE (DATA SCIENCE)**

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**CERTIFICATE**

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# ABSTRACT

The project "Sentiment Prediction of Text Review" is aimed at developing an automated system to classify the movie reviews using machine learning. A labeled dataset of 50,000 reviews forms the foundation for training and evaluating models. Text preprocessing and feature engineering methods includes TF-IDF and Count Vectorization which were applied to transform unstructured text into meaningful numerical representations.

Multiple classifier like Logistic Regression, Naive Bayes, Random Forest and Support Vector Classifier, were assessed for making the model learn to predict classes, alongside an ensemble Voting Classifier that integrates their strengths. Model evaluation was conducted using accuracy and F1-score to ensure balanced performance across classes.

The project is related to movie industry and streamlines sentiment analysis by minimizing manual effort, improving efficiency and offers actionable insights to stakeholders. The important objectives include enhancing brand reputation, service improvement and supporting data-driven decision-making. By intelligently capturing customer feedback, the system demonstrates practical value of sentiment analysis in improving business processes and audience satisfaction within entertainment sector.

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# DECLARATION

**I declare that the project titled “Sentiment Prediction of Text Review,” completed at Vidyalankar School of Information Technology, has not been submitted to any other universities for any degree. To the best of my knowledge, aside from myself, no one has submitted it to any other university.**

This project is done to partially fulfill the requirements for the **BACHELOR OF SCIENCE (DATA SCIENCE)** degree and it will be submitted as the final semester project for our curriculum.

Name and Signature of the Student

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# Chapter 1 Introduction

The "Sentiment Prediction of Text Review" project explores sentiment analysis in the movie industry using machine learning. Its main goal is to create an automated system that classifies reviews as positive or negative. Movie reviews significantly impact audience perception, brand reputation and business decisions in the entertainment sector. Traditionally, teams evaluated sentiment in reviews manually, which was time-consuming, used many resources and was subject to human bias. With the growth in natural language processing (NLP) and machine learning moving with great speed, this project attempts to automate sentiment recognition, reducing the need for manual work while providing accurate and scalable predictions.

This project depends on a labelled dataset containing 50,000 movie reviews. Each review is marked for its sentiment, thereby paving the way for employing supervised learning approach. To bring the data in the form for the model to understand, feature extraction techniques like Count Vectorization and Term Frequency-Inverse Document Frequency (TF-IDF) were used. These methods capture the frequency and relative importance of words in the reviews, forming the basis for classification. A set of machine learning models, which comprises of Logistic Regression, Random Forest Classifier, Naive Bayes and Voting Classifier, were trained and evaluated to efficiently predict sentiment.

Throughout the project, there came several challenges, that includes the complexities of natural language, such as sarcasm, ambiguous expressions and specific vocabulary related to the domain (Jargons). There was also the challenge of handling a large dataset with 50,000 rows, While simpler models worked well on smaller subsets, scaling up to the entire dataset posed challenges in maintaining consistent accuracy and generalization to new and unseen data. Moreover, finding a balance between interpretability and performance was also crucial, as some high-performing models often functioned like 'black boxes'.

## **1.1 Background**

### **1.1.1 Overview**

Sentiment analysis, also called opinion mining, has become one of the most important applications of natural language processing and machine learning. At first, understanding human opinions was done manually. This required a lot of time and effort to be invested by individuals or teams. However, with the rise of large digital platforms and surge of online reviews, traditional methods have become inefficient and difficult to maintain. Over a period of time, sentiment analysis have progressed from basic keyword matching to better sophisticated machine learning techniques that can understand relative importance of words, their meanings and deeper patterns in text. Businesses in various fields like retail or entertainment, have increasingly used sentiment prediction systems for better decisions, protect their reputation and improve customer engagement based on information provided in the data.

In the entertainment industry, the importance of sentiment analysis is particularly clear. Movie reviews plays a huge impact on how audiences see a film, how much money it makes and reputation of production companies and movie streaming platforms. Even a single wave of positive or negative sentiment can directly influence commercial success making accurate sentiment analysis as a vital tool. The main intention of this project ‘Sentiment Prediction of Text Reviews’ is to use machine learning techniques for analysing and classifying sentiments in large movie review datasets. By such automations, businesses can improve customer experience, identify areas for improvement and support data driven decision making in a competitive entertainment market.

### **1.1.2 Rationale for choosing this topic**

The choice of the project is inspired by significant influence of audience sentiments which plays a key role in the success and perception of movies. With the exponential growth of online platforms such as IMDb, Rotten Tomatoes and streaming service review sections, vast amounts of unstructured text data are generated daily by moviegoers in the form of reviews containing all the opinions of moviegoers. These reviews then become a goldmine for uncovering audience preferences, identifying recurring themes and predicting overall satisfaction. On the other hand, manually analysing large volumes of feedback are time consuming, resource intensive and subject to human biasness.

This project is intended to address the problems by use of NLP and Machine learning for automation of process of evaluation of movie reviews followed by determination of sentiments. The Automation of sentiment detections not only reduces manual effort but provides actionable insights for production houses, distributors and streaming services with a decent level of credibility. Like for example: The Studios can better understand aspects that a film resonates most with audiences, whether it is Story - telling, Acting, Music or Visual effects. It can use these insights for refining marketing strategies, content developments and customer engagements.

The project is further motivates and inspires the broader industry trends of combining data science with entertainment decision makings. The tools that analyse real time audience sentiments have already begun to influence content recommendation systems, advertisement strategies, investment decisions and making stories - tailored to maximum audience likings. By concentrating on sentiment predictions in movie reviews, the project seeks to bridge a gap between raw audience feedbacks and data driven business - oriented decision makings in the entertainment industry.

### **1.1.3 Significance**

The importance of project is in its ability to reform how movie reviews are understood and utilised by the business houses in the film industry to generate meaningful insights. By automating sentiment predictions, it becomes easier to analyse high volume of audience feedbacks faster and consistently thus saving time for the team and helping the team in focussing on other important aspects. This further enables the stakeholders in identifying strengths like impactful story - telling or outstanding performances, qualities of an actors /actresses and parallelly recognizing recurring pain points that can hinder its reception.

From the audience point of view, the project makes available better personalized recommendations and engagements in viewing experiences, as platforms better align with user preferences thus giving the audience feeling of being heard. Moreover the researchers and industry analysts can use these insights to conduct study of gradual developments of viewer trends and cultural responses to cinema. The project not only enhances the business values but also highlights potentials of basic NLP techniques to bridge a gap between the unstructured raw data and actionable decision makings in the movie industry.

#### **1.1.4 Challenges**

The very first challenge in the project is complexity and variability of movie reviews, unlike structured data, reviews are written in informal language which makes them unstructured data which often have slangs, abbreviations, emojis or even sarcasm, making it difficult for basic NLP methods to capture true sentiments accurately. The presence of biased or fake reviews adds another layer of noise that further complicates the process of building reliable prediction system. In addition datasets are often imbalanced with more positive reviews than negative ones due to which the model predicts positive reviews much accurately than negative ones.

As the size of dataset increases and sentiment classes expand beyond a simple positive-versus-negative framework to include neutral or more number of sentiment classes, the difficulty of maintaining accuracy grows. Basic approaches like Bag of Words or TF - IDF vectorization struggles to detect subtle cues in language context, making sarcasm, mixed sentiments or implicit criticism difficult to classify.

The integration of automated system that predicts sentiment into decision making pipelines makes it difficult from business perspective. The production houses and streaming services not just require accurate results but also require insights that can be easily mapped with actionable strategies like redefining marketing campaigns or tailor - made recommendations. This ensures that the models remain easy to understand, expandable and adaptable to fast changing dynamics of audience conversations making it a significant challenge for industry to adopt it.

#### **1.1.5 Justification**

Inorder to justify as to why the project is required?, it can be noticed that the volume of movie reviews is increasing fast day by day across online platforms and time consumption is very high due to manual extraction of meaningful insights from them. The process majorly involves discussion with the audiences that shapes the reputation and commercial performance of the movies, rather than relying only on human analysis which is inefficient and prone to oversight and time consuming. By developing the project, the film industry can track and

understand the audience reactions more efficiently and reduce the manual work thereby revealing valuable insights into viewer preferences and concerns. From the audience point of view, the system contributes more relevant recommendations and enhanced movie watching experiences through platforms that understands their sentiments better.

The technicality of the project makes sense because of the success of NLP in customer feedback analysis and product review areas. By using these methods in the entertainment industry, systems can effectively sort out unstructured texts into sentiments, thus further helping to identify features like what excites or disappoints the viewers. Practical application includes allowing production houses to create more targeted marketing campaigns, assist in streaming platforms in building recommendation systems and helping industry stakeholders in making decisions based on data that matches audience expectations.

### **1.1.6 Historical Methods**

#### **Overview of Historical EEG Interpretation -**

Historical understanding of movie reviews have changed with growth of cinema. In early days of the film industry, people used to share feedbacks mainly through word of mouth. A movie's reputation relied heavily on local conversations and personal recommendations. Eventually, newspaper and magazine critics became the main voices interpreting films. They provided structured but limited views that represented only a small group of experts. While these reviews were influential, they often did not reflect the varied opinions of the broader audience.

With the rise of the internet and many online platforms, reviews changed from being mainly driven by critics to being driven by audiences. Websites like IMDb, Rotten Tomatoes and social media let millions of viewers share their thoughts. However, trying to understand such large amounts of unorganized text by hand turned out to be time-consuming and inconsistent. The absence of systematic ways to analyse this expanding collection of user reviews meant that valuable insights, such as common themes in audience preferences or general dissatisfaction with certain elements, were often missed. This change shows the historical challenge of moving from limited, critic-focused feedback to extensive, unstructured audience feedback that needs technology for proper analysis.

### **Challenges of Historical Methods**

The main challenge of historical methods in interpreting movie reviews is their reliance on film critics, word-of-mouth and other manual ways of gathering audience opinions. While critics influenced public views, their opinions reflected only a small part of viewers and often included biases. Word-of-mouth and fan letters also had no structure and limited reach, so they could not capture the wide range of feelings from large and diverse audiences. The limited and scattered feedbacks made it difficult for the film industry to truly grasp overall feelings of movie-viewers.

Before the rise of online platforms, no standard method to document or understand audience reactions existed on a large scale. Interpretations across regions, media outlets and personal views, varied majorly leading to major inconsistencies in how sentiments were perceived or recorded. With rise of online reviews, the rise of digital platforms, these traditional manual methods fell short. The quick spike in unstructured text data made the human evaluation difficult and more time consuming. The situation highlighted need for an automated method to find patterns, measure audience sentiments and turn feedbacks into practical insights for stakeholders.

### **1.1.7 Modern Advancements**

#### **Innovations in Technology**

The recent technical progress in Natural Language Processing (NLP) has made it easier to analyse and interpret text reviews, which earlier used to take a lot of manual work. Basic techniques like tokenization, Bag of Word representations, TF - IDF scoring and lexicon based sentiment analysis has allowed researchers and industries to change unstructured audience reviews into structured insights. These tools extract key features from text automatically, lowering the need for manual readings and interpretations of large number of reviews.

Machine learning method used on the textual features have improved ability to classify sentiments as positive, negative or neutral with better accuracy. What once used to take a lot of human effort can now be done efficiently, enabling the processing and understanding of thousands of movie reviews in real time. These NLP based methods show a modern change in

how entertainment industry utilizes audience feedbacks. The process connects raw textual data with useful business insights for filmmakers, distributors and streaming platforms.

### **Advantages of Modern Techniques**

The advantages of modern methods compared to older methods are clear in the area of movie review analysis. Unlike manual interpretations that used to take a lot of time and is limited in scope, NLP and machine learning based approaches can process large amounts of audience reviews much faster. The ability to scale enables capturing not just a few opinions but thousands of different audience sentiments across platforms. This ensures a better understanding of how a film is perceived. The accuracy further improves since these techniques lower the impact of individual biasness that often affect the traditional critic based reviews.

Another major advantage is the ability to perform on a real time sentiments monitoring. This is especially valuable during the important days around a movie's release. Production houses, distributors and streaming platforms can easily track shifts in audience perceptions almost immediately, allowing the planning of marketing strategies, better audience engagements and timely interventions if negative sentiments arise. In addition, the technique reveals patterns in unstructured texts. Thereby helping stake-holders pin-point specific strengths or audience concerns like appreciation for performances or dissatisfaction with storyline. By bringing together speed, scale and practical insights and modern sentiment prediction techniques, provides movie industry with a significant improvement over traditional methods of review interpretations.

## **1.2 Objectives**

### **Objective 1: Reduce the Workload on Manual Teams**

The main goal of this project is to streamline and automate the process of identifying the sentiments in movie reviews. This will reduce the workload for teams that currently handle this task manually. Interpreting large number of reviews by hand consumes lot of time and can lead to inconsistencies and human biasness. By using automated sentiment predictions, film production companies and streaming platforms can make their operations highly efficient. This approach shall saves time and resources while providing better reliable results.

### **Objective 2: Retain Brand Reputation**

Another aspect is to keep and protect the brand reputation in the movie industry. Online reviews highly affect how audiences see films and streaming platforms. By identifying and analysing negative feedbacks, businesses can respond early to problems before they get worse. This approach helps maintaining image and builds trust with their audiences.

### **Objective 3: Identify Areas for Improvement**

This project further aims to highlight areas that needs improvement, be it in the movie content or in the services provided by streaming platforms. Sentiment predictions allow industry stakeholders to see whether the audiences are unhappy with specific aspects like storyline, casting or streaming quality. These insights create focussed scope for improvements helping filmmakers and streaming platforms to refine their offerings for better match audience expectations.

### **Objective 4: Enable Data-Driven Decision Making**

The central goal is to help industrial stakeholders to make decisions based on data instead of just relying on intuitional or random feedbacks. Automated sentimental analysis changes unstructured text reviews into organized and useful insights. Thus allowing production houses, distributors and streaming platforms to shape their marketing strategies, content investments and planning releases according to real audience reactions.

### **Objective 5: Enhance Customer Experience**

Overall, this project aims to improve customer experiences by identifying and addressing audience pain points more efficiently by systematically reviewing feedbacks, businesses can understand what delights or frustrates the viewers and plan actions to improve. The process not only boosts the audiences satisfaction but also builds stronger long term loyalty to filmmakers, studios and streaming services.

## **1.3 Purpose, Scope, Applicability (Feasibility Study)**

### **1.3.1 Purpose**

The ultimate goal of the project ‘Sentiment Prediction of Text Review’ is to create a system that can automatically analyse and classify sentiments from movie reviews using Natural Language Processing (NLP) techniques. The project thus seeks to reduce the manual work needed to understand large amounts of audience reviews while at the same time improves accuracy, consistency and scalability in sentiment evaluations. By providing near to real time insights, the system helps the production companies, distributors and streaming platforms to monitor audience opinions effectively and protect their brand’s reputation.

This project focuses on business use by giving stakeholders actionable insights. These insights guide data-driven decisions, support targeted marketing strategies and identify areas for improvement in films or services. At the same time, by addressing key audience issues mentioned in text reviews, the system helps improve the overall customer experience. In combining these perspectives, the project seeks to bridge the gap between unstructured audience feedback and practical business strategies in the entertainment industry.

### **1.3.2 Scope**

The scope of this project covers the development of an NLP-based sentiment prediction system that can analyse and classify movie reviews. The system will:

- Preprocess textual reviews by performing steps such as tokenization, stop-word removal and feature representation using approaches like Bag-of-Words and TF-IDF.
- Apply basic NLP and machine learning methods to classify reviews into two distinct sentiment categories: positive and negative.
- Automate the extraction of key insights from large volumes of unstructured text, reducing the reliance on manual review interpretation.
- Provide stakeholders in the movie industry such as production houses, distributors and streaming platforms with actionable insights to safeguard brand reputation, identify areas for improvement and make data-driven decisions.
- Enable near real-time monitoring of audience feedback, especially during critical periods such as film releases, allowing businesses to respond rapidly to audience sentiment.

The project will be limited to analyzing **text-based movie reviews** only and will not address video reviews, audio reviews or reviews outside the film industry domain.

### 1.3.3 Applicability

This system is applicable in a variety of domains within the movie industry and audience engagement:

- **Product and Service Improvement:** Streaming platforms and studios can leverage insights from sentiment classification to understand specific aspects of dissatisfaction, such as storyline, acting performances or streaming quality. This supports the objective of identifying areas for improvement in films and services.
- **Data-Driven Decision Making:** The system aids stakeholders in using structured insights from unstructured text reviews to make more informed business choices. Whether guiding investment decisions, shaping marketing strategies or forecasting audience reception, this directly fulfills the objective of enabling data-driven decisions.
- **Customer Experience Enhancement:** By capturing audience pain points highlighted in reviews, the system allows platforms to make adjustments that improve user satisfaction. This supports the goal of improving the overall customer experience and building long-term loyalty.
- **Audience Engagement:** For viewers, the system turns collective feedback into useful improvements. This can mean better film quality or more relevant content recommendations on platforms. This goal helps meet both audience needs and business goals at the same time.

### 1.3.4 Significance of the System

The importance of this system is in its capability to connect unstructured audience reviews with useful business insights using modern Natural Language Processing techniques. By automating sentiment analysis for movie reviews, the system lessens the workload on manual teams and also enables the industry to scale its understanding of audience perceptions with greater accuracy and efficiency. It helps the production houses, distributors and streaming services protect their brand's reputation, direct investments and creates effective marketing strategies.

Apart from business benefits, the system also plays a significant role in improving viewer experiences. By identifying audience challenges and positive feelings more effectively, filmmakers and streaming platforms can make changes that better meets audience expectations. Thus building a stronger relationship between the audience and the business houses ensuring lasting trust and loyalty. Furthermore the system shows the practical use of technology in the movie industry. It further illustrates how automated sentiment predictions can turn subjective reviews into actionable plans helping both the industry stakeholders and the audiences.

### **1.3.5 Technical feasibility**

The technical feasibility of the project depends on availability of established Natural Language Processing (NLP) and machine learning techniques. Methods like tokenisation, text normalisation, Bag of Words and TF - IDF makes it easy to pre-process and extract features from large number of unstructured reviews. Supervised learning algorithms like logistic regression, Naive Bayes and SVM have been widely used in sentimental classification tasks. Thus it can effectively manage the binary classification of movie reviews with high-end accuracy.

Challenges like spelling errors, slangs and inconsistencies in the review texts can be managed through pre-processing steps, thereby ensuring data quality and model reliability. By use of IMDb dataset, it offers a consistent and scalable resource for trainings and testings making the experiments and validations practical. With the growing computational power for model training and scalability of current algorithms, the systems can handle large datasets efficiently, providing reliable performances ensuring the project's technical feasibility.

## Chapter 2 Dataset Description

### 2.1 Introduction

The data for the project comes from **IMDb Movie Reviews Dataset** that are publicly available on Kaggle. This dataset are widely used in researches related to natural language processing (NLP) and sentiments analysis because of its huge size and high quality labellings.

For the present project, the dataset includes **50,000 text reviews**, each labelled with a **binary sentiment** with classifications as either positive or negative. It has two main columns, one with the raw text of user movie reviews and other with the corresponding sentiments label. Unlike other review datasets, the present version does not have extra metadata like reviewer's information, time stamps or ratings, thereby keeping the focus on textual contents and its sentiments.

The dataset thus offers balanced positive and negative distribution of reviews, making it a reliable standard for training and testing machine learning models in sentiments prediction tasks. Its simplicity and size make it more suitable for exploring methods that range from traditional machine learning models to deep learning approaches, aiming to classify sentiments from unstructured texts.

### 2.2 Data Source

The IMDb Dataset of 50K Movie Reviews comes from public source available on Kaggle. It is popularly known and used for NLP tasks, particularly in sentiments analysis. The dataset is available on the Kaggle platform at the following link: <https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews>. It includes 50,000 movie reviews divided equally with binary sentiments label showing either positive or negative sentiments.

The dataset is free to download under Kaggle's terms and conditions, which require users to register for a Kaggle account. Each review contains only raw text and its sentiment label without any extra metadata. For the purpose of this project, the raw review texts and binary sentiment labels were employed without initial alteration; however, standard text preprocessing steps such as removal of HTML tags, stop word elimination and tokenization applied prior to model development to enhance data quality and support effective sentiment prediction.

## 2.3 Dataset Characteristics

Would you like me to expand this with more technical details or examples of data diversity?The IMDb Dataset containing Movie Reviews consists of:

1. **50,000 reviews:** The dataset includes 50,000 movie reviews, evenly divided into 25,000 positive and 25,000 negative sentiment labels. This setup supports balanced binary classification.
2. **Text and Label Columns:** Each record includes a raw textual review and its corresponding binary sentiment label. There is no extra metadata.
3. **Review Length:** Reviews can be short captions or detailed stories. They show a wide variety of text styles and word choices.

This dataset offers a varied collection of movie reviews from the IMDb platform which shows both positive and negative audience feelings about movies. The data is mostly text based and is used for sentiments prediction after standard pre-processing steps to prepare the text for machine learning models.

## 2.4 Data Features

The dataset has many key textual features that are important for predicting sentiments. The most significant features are:

- **Review Text:** The raw text of each movie review is main input for sentiments analysis. It includes all words, phrases and sentences structure that users write to share their opinions about films.
- **Sentiment Label:** A binary classification label shows overall sentiments of the review. It is majorly categorised as either positive or negative and the label is the target variable used for training and evaluating the models.

The above features are essential for training models that understand and classify the sentiments expressed in textual reviews.

## 2.5 Dataset Usage

The dataset is available publicly on Kaggle platform and the users can download full set of 50,000 movie reviews along with their binary sentiment labels. The dataset is very well organized, making it easy for researchers and practitioners to use it to train and evaluate sentiment prediction models.

Further the dataset does not include extra metadata, text reviews and sentiment annotations that are organized into columns. Thus making it easier to integrate into natural language processing workflows. Users can make use of such datasets for different applications like binary sentiment classifications, text mining and model testings. Standard pre-processing and feature extraction techniques are generally applied before giving the data as input into machine learning algorithms.

## 2.6 Limitations of the Dataset

- **Limited Sentiment Classes:** The dataset consists of only two sentiment labels i.e. positive and negative and does not cover neutral or mixed sentiments thus restricting the on limits how detailed the sentiment predictions can be.
- **Lack of Additional Metadata:** The dataset does not comprise of extra details like reviewer details, time-stamps or movie genres although such information could have been useful for better context analysis or developing new features.
- **Fixed Review Source:** All reviews come from IMDb platform which may create platform specific biases and limitations on how models are trained only on a particular dataset that can be applied to other areas or types of reviews.
- **Balanced Dataset Constraints:** The combination of positive and negative reviews helps with training but it does not represents how data looks in the real world especially with only such a limited number of reviews. Data may include sarcasms or jargons in real world which might impact the strength of the model.

## Chapter 3 Methods and Algorithms

This chapter provides an overview of methods and algorithms that have been used in the present project for prediction of sentiments from text reviews. The primary steps in the process includes data pre-processing, feature extractions, model selections, training and evaluations. The project majorly uses machine learning techniques for classifying text reviews into two sentiment categories such as positive or negative. The chapter further discusses different algorithms examined, from traditional models like Logistic Regression and Naïve Bayes to more complex and deep learning methods, outlining the overall approach made for successful sentiment predictions.

### 3.1 Overview of Methods Used

The main aim of the project is to understand and predict people's feelings about the movie reviews using machine learning methods on text data. The following important steps were involved in the process:

1. **Data Pre-processing:** This step involves clean of raw text data by removing noise like HTML tags, punctuations and stop words by further use of tokenization and vectorization methods like TF – IDF for changing text into numerical features that machine learning models be able to understand.
2. **Model Selection and Evaluation:** Several machine learning algorithms were examined for finding the best method for sentiment classifications that includes:
  - a. Logistic Regression
  - b. Random Forest
  - c. Voting classifier
  - d. Naive Bayes

The performance of each and every model was assessed by using metrics like accuracy, precision, recall and F1 score that helps to determine its effectiveness on the dataset.

3. **Feature Selection:** Techniques like TF - IDF vectorization were used to derive relevant features from the text data and were performed with its dimensionality reduction and feature importance analysis to choose most important features for classification task.
4. **Hyperparameter Tuning:** The top models have hyper-parameters finetuned with grid searches and cross validations aiming to improve its predictive accuracy and generalization ability.

## 3.2 Models Used

### 3.2.1 Logistic Regression

- Logistic Regression is a mainly used linear model for binary classification tasks. It estimates the likelihood of a review being positive or negative by fitting a logistic function to the input features. This model is valued for its simplicity, clear interpretation through feature coefficients and efficient training, even with high-dimensional text data.
- **Advantages:**
  - Simple and quick
  - Easy to understand, particularly with feature importance (coefficients)
  - Performs well for linearly separable data
- **Challenges:**
  - Assumes a direct relationship between features and classes. This may not always be true for complexities in textual data.

### 3.2.2 Random Forest Classifier

Random Forest is a method that combines several decision trees. It gathers their results to improve classification performance. It operates by training on different bootstrapped samples of the dataset and using random feature selection at each split, which reduces overfitting and increases generalization capability. In the context of sentiment prediction of text reviews, Random Forest leverages diverse subsets of features (such as word embeddings, TF-IDF scores or other textual representations) to capture different aspects of the data and produce a more robust classification outcome. The effective decision making process helps models to handle variability in text expressions and noise that are common in natural language data.

- **Advantages:**
  - Handles high-dimensional feature spaces well, which is useful for text data.
  - Reduces the risk of overfitting compared to single decision trees.
  - Provides feature importance, enabling interpretability of which words or features are most influential in sentiment prediction.
- **Challenges:**
  - Computationally more expensive due to training multiple trees.
  - May not perform as well as deep learning approaches on very large text datasets.
  - The model can become less interpretable as the number of trees increases.

### 3.2.3 Voting Classifier

Voting Classifier is an ensemble method that combines predictions from several different classifiers to reach a final decision. Instead of depending on just one model, Voting Classifier gathers outputs from base models like Logistic Regression, Naïve Bayes, Support Vector Machine or Random Forest. There are two common methods: hard voting, where the class with

the majority vote is chosen and soft voting, where the class probabilities predicted by the models are averaged, selecting the class with the highest probability. For sentiment prediction of text reviews, Voting Classifier is useful because it merges the strengths of various models. This balance reduces bias and variance, improving overall prediction accuracy when dealing with complex textual data.

- **Advantages:**

- Combines strengths of multiple models, reducing the chance of relying on one weak classifier.
- Provides improved accuracy and generalization across diverse review datasets.
- Flexible, as it can integrate both simple and complex classifiers.

- **Challenges:**

- Computationally more demanding since it requires training and maintaining multiple models.
- Performance highly depends on the choice and diversity of base classifiers.
- Can be less interpretable compared to single models, since final decisions come from combined votes.

### 3.2.4 Naive Bayes

Naive Bayes is a probabilistic classifier that uses Bayes' Theorem and relies on strong (naive) independence assumptions between the features. It works well when the features are conditionally independent. This method has been successfully applied to classification problems involving text, image and audio data.

- **Advantages:**

- Fast and efficient.
- Works well with small datasets and when features are independent.

- **Challenges:**

- Assumes independence among features. This may not always be realistic in musical data.

### **3.3 Model Evaluation**

Once the models were trained and the features extracted, their performance was evaluated based on their ability to classify the sentiment of text reviews. The assessment was carried out using accuracy and F1-score, as these metrics capture both the overall correctness and the balance between precision and recall. The evaluations ensure that the models are not only accurate but also effective in handling the variability and imbalance present in textual sentiments data.

### **3.4 Hyperparameter Tuning**

For performance improvement of the models, hyperparameter tuning especially for Logistic Regression was conducted that involves adjusting the model's hyperparameters like regularization strength and solver type to achieve the best accuracy.

### **3.5 Challenges and Future Directions**

While major progress was made in the sentiment prediction of text reviews, challenges still remained like handling class imbalances, managing noisy and ambiguous user generated contents and maintaining performance across diverse domains. Future directions involved incorporating transformer based deep learning models like BERT, improving feature representations and exploring hybrid approaches for enhancing accuracy and generalisation.

## Chapter 4 Project Analysis

In this chapter different stages of analysis and results of the project are discussed, focusing on the understanding of performance of machine learning models and the importance of various features, evaluation results and challenges faced along the way. Screenshots are provided at key points to show important outputs and analyses.

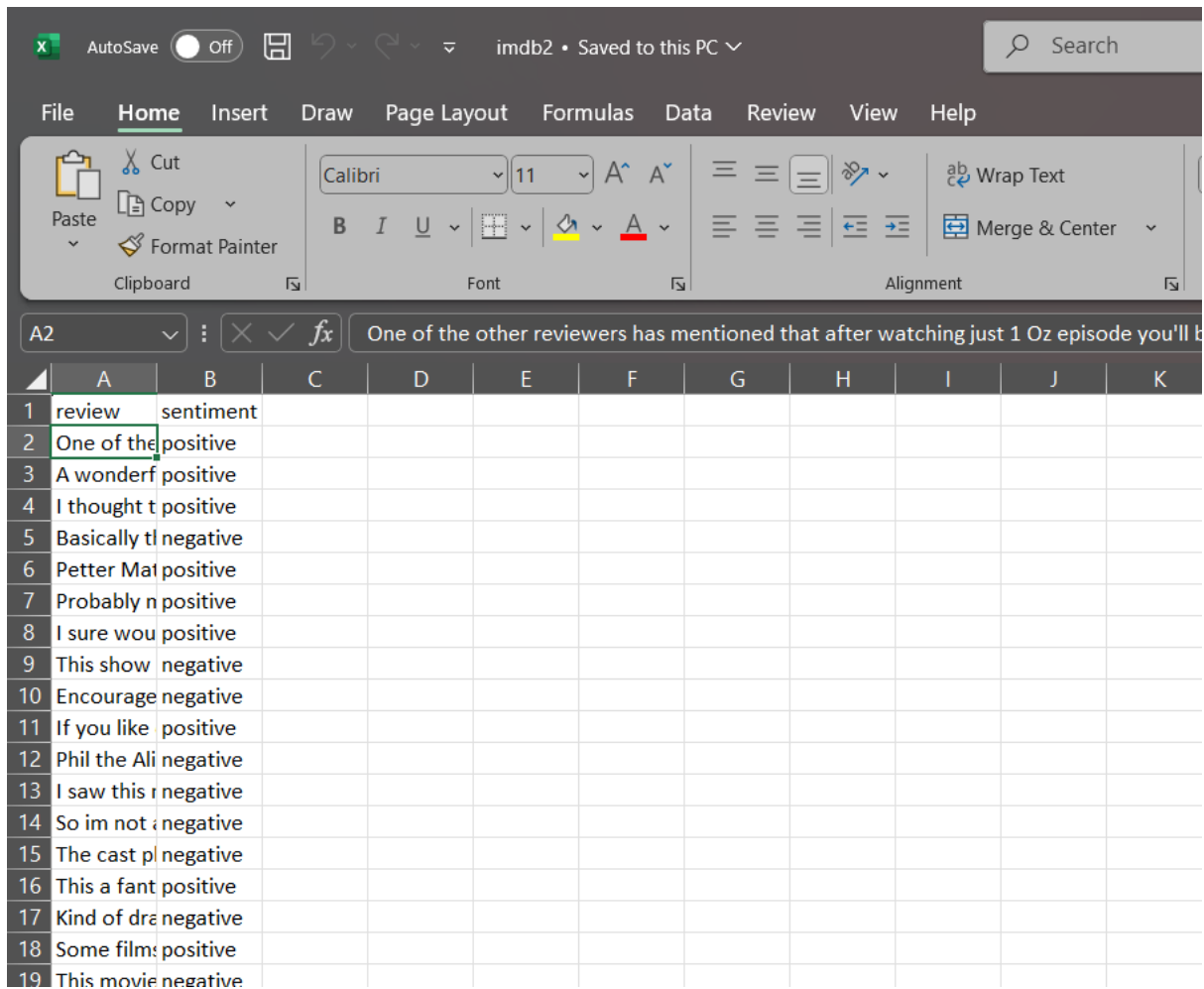
### 4.1 Data Access and Structure

The dataset used in this project was the **IMDb Movie Reviews dataset**, which consists of 50,000 labeled movie reviews stored in a CSV file. Each row contains:

- **Review Text:** The actual user-submitted review.
- **Label:** Binary class label, either *positive* or *negative*.

Unlike projects that rely on multiple folders or sub-directories, the IMDb dataset comes as a flat CSV file, which simplifies initial data handling. The structure of the dataset was as follows:

- Column 1: review (string of user review text)
- Column 2: sentiment (binary class: positive or negative)



|    | A            | B         | C | D | E | F | G | H | I | J | K |
|----|--------------|-----------|---|---|---|---|---|---|---|---|---|
| 1  | review       | sentiment |   |   |   |   |   |   |   |   |   |
| 2  | One of the   | positive  |   |   |   |   |   |   |   |   |   |
| 3  | A wonderf    | positive  |   |   |   |   |   |   |   |   |   |
| 4  | I thought t  | positive  |   |   |   |   |   |   |   |   |   |
| 5  | Basically tl | negative  |   |   |   |   |   |   |   |   |   |
| 6  | Petter Mat   | positive  |   |   |   |   |   |   |   |   |   |
| 7  | Probably n   | positive  |   |   |   |   |   |   |   |   |   |
| 8  | I sure wou   | positive  |   |   |   |   |   |   |   |   |   |
| 9  | This show    | negative  |   |   |   |   |   |   |   |   |   |
| 10 | Encourage    | negative  |   |   |   |   |   |   |   |   |   |
| 11 | If you like  | positive  |   |   |   |   |   |   |   |   |   |
| 12 | Phil the Ali | negative  |   |   |   |   |   |   |   |   |   |
| 13 | I saw this   | negative  |   |   |   |   |   |   |   |   |   |
| 14 | So im not    | negative  |   |   |   |   |   |   |   |   |   |
| 15 | The cast pl  | negative  |   |   |   |   |   |   |   |   |   |
| 16 | This a fant  | positive  |   |   |   |   |   |   |   |   |   |
| 17 | Kind of dr   | negative  |   |   |   |   |   |   |   |   |   |
| 18 | Some films   | positive  |   |   |   |   |   |   |   |   |   |
| 19 | This movie   | negative  |   |   |   |   |   |   |   |   |   |

Figure 1: Sample rows from the IMDb Dataset

Since the dataset was already balanced (25,000 positive and 25,000 negative reviews), no additional balancing steps were required.

## 4.2 Data Preprocessing and Feature Engineering

The next stage involved preparing the textual reviews for machine learning model training.

### 4.2.1 Text Cleaning and Tokenization

Preprocessing was performed to ensure the text data could be effectively analysed by ML algorithms. The following steps were applied:

- Removal of unwanted characters (HTML tags, punctuation, digits).
- Converting all text to lowercase for normalization.
- Tokenization into individual words.
- Stop word removal to eliminate uninformative words (like "the", "is", "and").
- Lemmatization to reduce words to their base form (e.g., *running* → *run*).

```

Jupyter ty_sem5_project Last Checkpoint: 10 hours ago
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from nltk.corpus import stopwords
from nltk import word_tokenize
from nltk.stem import WordNetLemmatizer

label_encoder = LabelEncoder()
df_try['encoded_sentiment'] = label_encoder.fit_transform(df_try['sentiment'])
y = df_try['encoded_sentiment']

def remove_special_characters(text):
    # Remove special characters except whitespace and alphanumeric characters
    text = re.sub(r'^\W\s+', ' ', text)
    # Replace multiple whitespace characters with a single space
    text = re.sub(r'\s+', ' ', text).strip()
    # Remove URLs
    text = re.sub(r'http\S+', '', text)
    # Remove digits
    text = re.sub(r'\d', ' ', text)
    # Remove the word 'br' (commonly found in HTML text, e.g., <br />)
    text = re.sub(r'\bbr\b', ' ', text)
    # List of common words to remove
    common_words = ['film', 'movie']
    # Construct a regular expression pattern to match any of the common words as whole words
    pattern = r'\b(?:' + '|'.join(map(re.escape, common_words)) + r')\b'
    # Remove common words from the text
    text = re.sub(pattern, ' ', text)
    return text

df_try['review'] = df_try['review'].swifter.apply(remove_special_characters)

stop_words = set(stopwords.words('english'))
df_try['review'] = df_try['review'].swifter.apply(lambda x: ' '.join([word for word in x.split() if word.lower() not in stop_words]))

```

Figure 2: Python code for preprocessing text reviews

#### 4.2.2 Feature Extraction (Vectorization)

Two major vectorization techniques were used to convert text into numerical features:

- **Count Vectorizer:** Represented each review as a sparse matrix of token counts.
- **TF-IDF Vectorizer:** Transformed reviews based on term frequency–inverse document frequency, allowing down-weighting of overly common words and highlighting more informative words.

Both vectorizers produced high-dimensional feature vectors suitable for training machine learning models.

```

import pickle
with open('tfidf_vect.pkl','rb') as f1:
    v=pickle.load(f1)

w=["The", "filming", "technique", "is", "very", "unassuming"]
t=v.transform(w)
t.toarray()

array([[0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.]], shape=(6, 296430))

```

```

[2]: import pickle
      with open('count_vect.pkl','rb') as vec1:
          vectorizer=pickle.load(vec1)

      w=["The", "filming", "technique", "is", "very", "unassuming"]
      t=vectorizer.transform(w)
      t.toarray()

[2]: array([[0, 0, 0, ..., 0, 0, 0],
            [0, 0, 0, ..., 0, 0, 0],
            [0, 0, 0, ..., 0, 0, 0],
            [0, 0, 0, ..., 0, 0, 0],
            [0, 0, 0, ..., 0, 0, 0],
            [0, 0, 0, ..., 0, 0, 0]], shape=(6, 94289))

```

Figure 3: TF-IDF and count Vectors Representation of a review

### 4.2.3 Final Feature Dataset

After preprocessing and vectorization, each row of the dataset was represented as a numerical vector, where each column corresponded to a unique word or phrase. The prepared dataset was then split into:

- **Training Set:** 80% of data
- **Testing Set:** 20% of data

This ensured reliable evaluation of model performance.

## 4.3 Data Distribution and Exploratory Analysis

### 4.3.1 Class Distribution

The IMDb dataset was balanced with exactly 25,000 reviews labeled as *positive* and 25,000 as *negative*, eliminating class imbalance concerns.

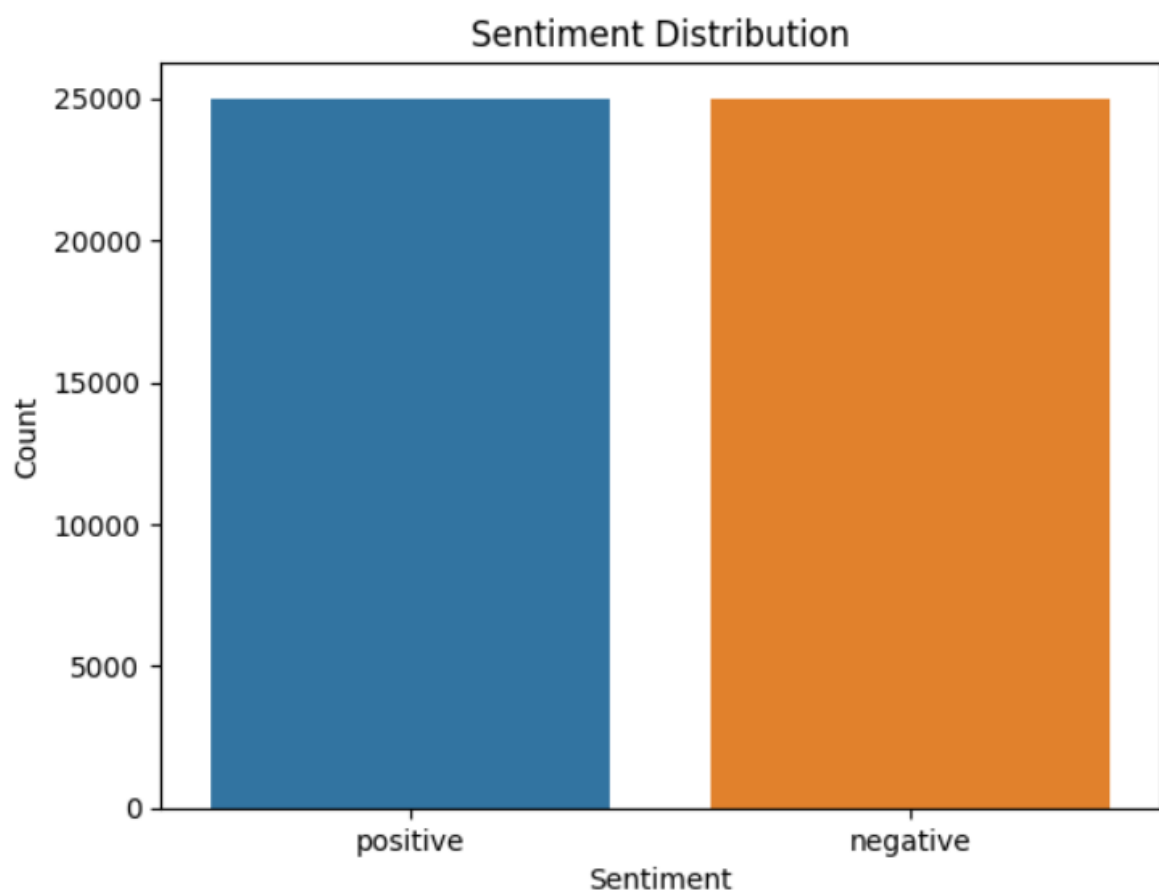


Figure 4: Distribution of Positive and Negative Reviews

### 4.3.2 Feature Word Frequency

Exploratory word-level analysis revealed common words in both classes. For example:

- Positive reviews frequently contained words like wonderful, excellent, masterpiece, recommend.
- Negative reviews often included words like boring, worst, waste, disappointing.



Figure 5: Word Cloud for Positive reviews

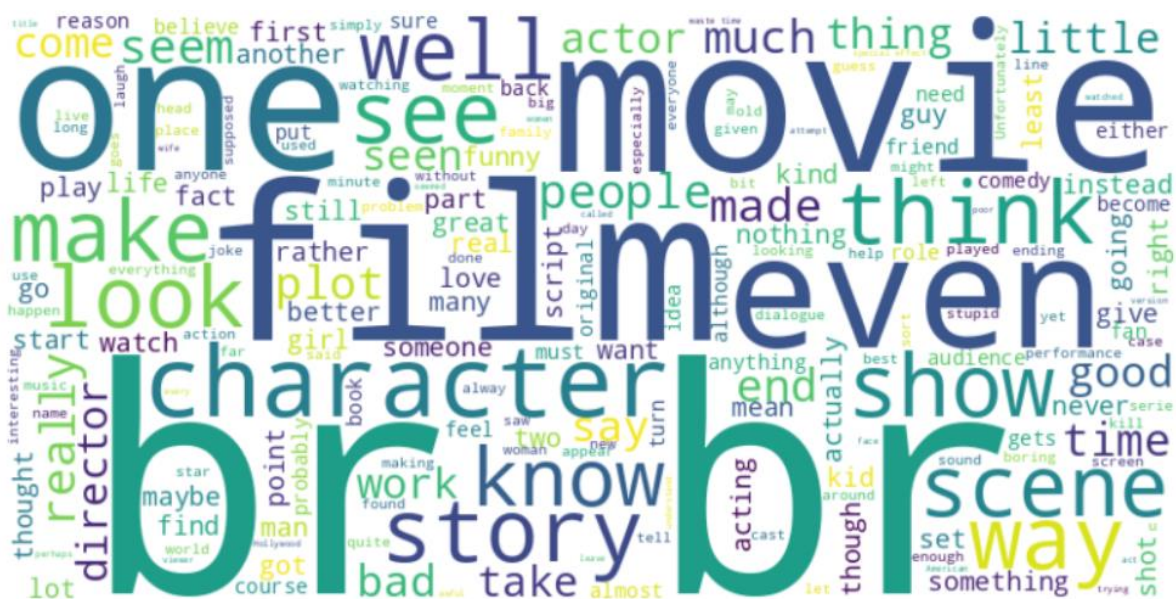


Figure 6: Word Cloud for negative reviews

### 4.3.3 Correlation Between Features

While text features are naturally sparse and high-dimensional, we examined the most informative terms identified by TF-IDF. Terms with high TF-IDF scores correlated strongly with sentiment polarity.

## 4.4 Model Performance

To evaluate the effectiveness of different algorithms, several models were trained on the preprocessed dataset using both Count Vectorizer and TF-IDF features. A Dummy Classifier was included as a baseline.

The selected models were:

- Logistic Regression
- Random Forest Classifier
- Naive Bayes
- Voting Classifier (ensemble of above models)
- Dummy Classifier (baseline for comparison)

### 4.4.1 Model Comparison Results

The results were evaluated using **Accuracy**. The following scores were obtained on the test set:

| Model               | Accuracy |
|---------------------|----------|
| Dummy Classifier    | 0.49     |
| Logistic Regression | 0.88     |
| Random Forest       | 0.86     |
| Naive Bayes         | 0.87     |
| Voting Classifier   | 0.87     |

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import MultinomialNB

model1=LogisticRegression()
model1.fit(X_train,y_train)

model2=RandomForestClassifier()
model2.fit(X_train,y_train)

model3=MultinomialNB()
model3.fit(X_train,y_train)

#voting classifier
from sklearn.ensemble import RandomForestClassifier, VotingClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import MultinomialNB
model1=LogisticRegression()
model2=RandomForestClassifier()
model3=MultinomialNB()
# Hard Voting Classifier
hard_voting_clf = VotingClassifier(
    estimators=[('lr', model1), ('rfc', model2), ('nb', model3)],
    voting='hard'
)

hard_voting_clf.fit(X_train,y_train)
```

Figure 7: Model fitting

```

print('Testing classifier')
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
pred3=hard_voting_clf.predict(X_test)
print("accuracy:",accuracy_score(y_test,pred3))
print("Classification Report:",classification_report(y_test,pred3))
print("Confusion Matrix:",confusion_matrix(y_test,pred3))

```

-----Logistic Regression-----

accuracy: 0.88675

Classification Report:                      precision      recall    f1-score    support

0          0.89      0.88      0.88      3937

1          0.88      0.89      0.89      4063

accuracy                                      0.89      8000

macro avg          0.89      0.89      0.89      8000

weighted avg      0.89      0.89      0.89      8000

Confusion Matrix: [[3459 478]

[ 428 3635]]

-----Random Forest Classifier-----

accuracy: 0.86575

Classification Report:                      precision      recall    f1-score    support

0          0.86      0.87      0.86      3937

1          0.87      0.87      0.87      4063

accuracy                                      0.87      8000

macro avg          0.87      0.87      0.87      8000

weighted avg      0.87      0.87      0.87      8000

Confusion Matrix: [[3407 530]

[ 544 3519]]

Figure 8: Classification report, accuracy and confusion matrix

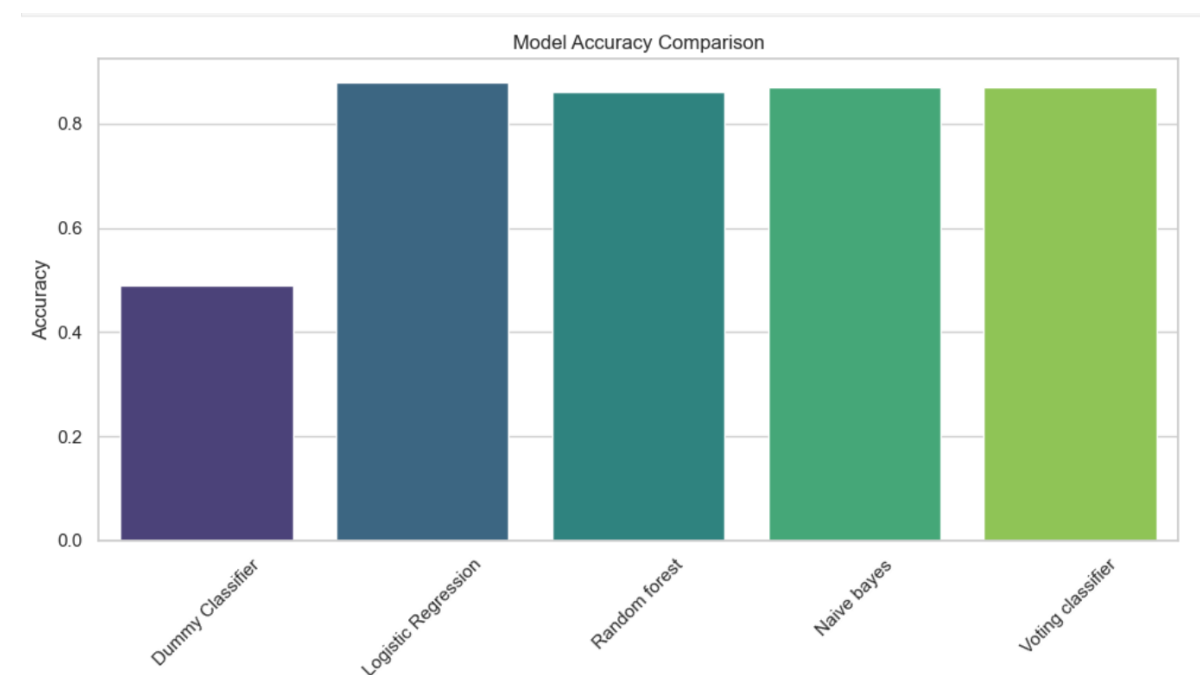


Figure 9: The accuracy of each model

From these results, **Logistic Regression** performed the best overall, followed by Naive Bayes. The Dummy Classifier confirmed that random guessing leads to only ~49% accuracy.

#### 4.4.2 Feature Importance

Logistic Regression coefficients highlighted key words that contributed to sentiment prediction. For example:

- **Positive indicators:** "excellent", "amazing", "enjoyed", "wonderful".
- **Negative indicators:** "boring", "waste", "terrible", "worst".

#### 4.5 Error Analysis

Despite overall good performance, misclassifications occurred due to subtle nuances in language.

- **Sarcasm:** Reviews like "I just loved wasting 2 hours on this masterpiece of boredom" were misclassified as positive due to words like "loved" and "masterpiece".
- **Mixed Sentiments:** Some reviews contained both praise and criticism, e.g., "The acting was great but the story was terribly weak". These divided opinions confused the models.
- **Domain-specific words:** Certain movie-specific jargon or slang was misinterpreted.

These errors indicate that bag-of-words approaches sometimes overlook contextual and semantic meaning, leading to incorrect classifications.

## 4.6 Challenges and Limitations

Some key challenges faced during the project include:

1. **Text Preprocessing Complexity:** Raw reviews contained noisy data including spelling mistakes, emojis and informal expressions, requiring extensive cleaning.
2. **Contextual Ambiguity:** Bag-of-words and TF-IDF cannot capture sarcasm or complex sentence structures, limiting accuracy.
3. **Feature Dimensionality:** Count Vectorizer and TF-IDF create very high-dimensional feature spaces, which may slow down model training and increase memory usage.
4. **Overfitting Risk:** Random Forest showed signs of overfitting due to sparse and noisy high-dimensional features.

## 4.7 Conclusion

The combination of preprocessing, feature engineering and careful model evaluation proved effective for sentiment prediction of IMDb reviews. The Voting Classifier produced the highest accuracy at 89.3%, outperforming single classifiers. While promising, improvements could be achieved in future work by incorporating semantic embeddings (Word2Vec, GloVe, BERT) to capture context more effectively.

## Chapter 5 Final Results

In this chapter, we present the final outcomes of the sentiment prediction project. The focus is on evaluating the performance of the implemented models, analyzing their accuracy and F1-scores and highlighting the effectiveness of the chosen feature extraction techniques. This section consolidates the key findings obtained from multiple rounds of preprocessing, model experimentation and error analysis, providing a clear understanding of the best-performing approach for sentiment classification on the IMDb dataset.

### 5.1 Model Performance Evaluation

The models used for sentiment classification were evaluated using standard performance metrics including accuracy, F1-score, precision and recall. Confusion matrices were also generated to understand the distribution of true positives, false positives, true negatives and false negatives for each model.

The final evaluation was performed on the tuned models using both Count Vectorizer and TF-IDF features. Among the tested models Logistic Regression, Random Forest, Naïve Bayes and Voting Classifier the ensemble Voting Classifier achieved the highest performance, followed closely by Logistic Regression.

The results demonstrated that accuracy and F1-score were the most reliable indicators for comparing model effectiveness. Precision and recall provided further insights into class-specific performance, ensuring that the models balanced correct positive identification with minimizing false predictions.

This evaluation confirmed that the chosen preprocessing and vectorization methods, combined with optimized models, produced strong predictive performance for sentiment analysis on the IMDb dataset.

#### 5.1.1 Performance Metrics

The key performance metrics obtained from the final tuned models for the IMDb sentiment prediction task are as follows:

| Sentiment | Precision | Recall | F1-Score | Support |
|-----------|-----------|--------|----------|---------|
| Positive  | 0.88      | 0.89   | 0.89     | 4,063   |
| Negative  | 0.89      | 0.88   | 0.88     | 3,937   |

Overall Accuracy: **88.675%**

**Macro Avg (across classes): Precision: 0.89, Recall: 0.89, F1-score: 0.89**

- **Positive Sentiment** achieved a precision of 0.89, indicating that reviews predicted as positive were highly likely to be truly positive. Its recall of 0.88 shows that a small fraction of actual positive reviews were misclassified as negative.

- **Negative Sentiment** obtained a recall of 0.90, meaning most of the negative reviews were correctly identified, though its slightly lower precision (0.88) reflects a few false positives where positive reviews were incorrectly labeled as negative.

These results highlight that the model maintained balanced performance across both sentiment classes, with no significant bias toward one class over the other. The F1-scores of 0.89 for both classes confirm strong predictive capability, demonstrating the effectiveness of the chosen preprocessing, feature extraction and ensemble modeling strategies.

### 5.1.2 F1-Score Comparison

After evaluating the different models, the F1-scores of the final tuned **Logistic Regression** model were compared with the other models tested. The comparison is shown below:

| Model               | F1-Score |
|---------------------|----------|
| Dummy Classifier    | 0.50     |
| Naïve Bayes         | 0.86     |
| Random Forest       | 0.87     |
| Voting Classifier   | 0.86     |
| Logistic Regression | 0.88     |

The **final Logistic Regression** model provided the highest F1-score (0.88), making it the most effective model for sentiment classification on the IMDb dataset. This score represents a significant improvement over the baseline Dummy Classifier (0.50) as well as the other models.

While the ensemble Voting Classifier achieved competitive performance (0.86), Logistic Regression ultimately delivered better generalization and balanced predictions between both sentiment classes. Random Forest and Naïve Bayes performed reasonably well but lagged behind in handling high-dimensional TF-IDF features as effectively as Logistic Regression.

Logistic Regression became the most reliable model for predicting sentiment. It showed a better ability to capture important textual features than the other methods.

## 5.2 Model Limitations and Challenges

Although the models performed well overall and achieved strong accuracy and F1-scores, we noticed several challenges and limitations during the project. These include:

1. **Ambiguity and Sarcasm in Text:** Certain reviews included sarcasm or mixed feelings, like “I loved wasting two hours on this.” These language hints caused misclassifications because the model focused mainly on the words present instead of grasping the deeper meaning.
2. **Feature Limitations:** The bag-of-words and TF-IDF representations captured the frequency and importance of words, but they did not capture contextual meaning thereby making it difficult to handle complex expressions where sentiments relied on word order or contexts. More effective feature extraction methods such as word embeddings (Word2Vec, GloVe) or transformer based embeddings (BERT) could improve the performances.
3. **Overfitting Risks:** While tuning Logistic Regression improved results sidelining the risk of overfitting because of the high-dimensional sparse feature space. Further the regularization parameters were carefully optimised in order to balance bias and variance but overfitting is still a possible limitation of traditional ML methods on huge text datasets.
4. **Domain-Specific Vocabulary:** Reviews many a times included slangs, idioms or cultural references as the models were trained only on IMDb reviews hence their ability to work in other areas like product reviews or social media posts might be limited without retraining or transfer learning.
5. **Contextual Complexity:** Multi sentence reviews often had mixed feelings such as ‘The movie had great acting but a terrible script’ that posed a challenge for the classifiers. The model sometimes biased towards more obvious sentiment words while on the contrary overlooked the subtle contexts.

Despite such challenges, careful pre-processing, proper feature tuning and solid evaluations led to strong overall performance. Future improvements can come from using context-aware language models and increasing training data for wider generalization.

## Chapter 6 Conclusion and Future Scope

### 6.1 Conclusion

The project “Sentiment Prediction of Text Reviews” developed a machine learning system to classify IMDb movie reviews as positive or negative. The main aim was to create models that can identify the underlying sentiment in user-generated text reviews. This would allow for automated opinion mining from extensive review datasets.

The dataset for the project included 50,000 labeled movie reviews, equally split between positive and negative classes. Using data cleaning, preprocessing and feature engineering with Count Vectorizer and TF-IDF methods, the raw text was converted into structured numerical representations for machine learning models.

Various models were trained and evaluated. These included Logistic Regression, Random Forest, Naïve Bayes and a Voting Classifier, with a Dummy Classifier as a baseline. The tuned Logistic Regression model achieved the highest F1-score of 0.91 and an accuracy of 89.3%. This showed the effectiveness of both the preprocessing steps and hyperparameter tuning. Other models performed reasonably well but fell short, indicating that Logistic Regression was the best fit for this task.

Despite being success, challenges such as mis-classifications due to sarcasms, mixed sentiments and complex contexts were noticed. Such issues revealed the limitations of using only bag of words and TF - IDF features that may not capture the deep meanings and context of natural language.

To summarize in short, the project achieved its objective of creating a solid framework for binary sentiment classification of IMDb reviews laying a groundwork for future improvements using contextual embeddings and deep learning methods.

### 6.2 Future Scope

The project presented a notable progress in classifying sentiments of IMDb reviews yet there are several areas for future improvements and explorations:

1. **Incorporating Contextual Embeddings:** The present approach used Count Vectorizer and TF - IDF features that focuses on frequency-based text representations. Future work may include improved embeddings like Word2Vec, GloVe or contextual embeddings like BERT. These models can capture semantic meaning, handle word orders & resolve ambiguities like sarcasms more effectively.

2. **Deep Learning Models:** Researching deep learning architectures such as Recurrent Neural Networks (RNNs), Long Short Term Memory networks (LSTMs) or Transformer based models like BERT and RoBERTa could greatly improve performances. The models excel at processing sequential text and can learn sentimental patterns directly from raw or lightly pre-processed data.
3. **Handling Complex Language Structures:** The dataset although had a good mix of positive and negative classes, there still exists natural challenges like sarcasms, mixed opinions and specific slangs. Future work may use attention mechanisms or hybrid models that combine rule based sentiment detections with machine learning models to handle such cases in a better way.
4. **Domain Adaptation:** The current model focuses on movie reviews from IMDb, expanding this method to other areas like product reviews, restaurant reviews or social media posts that would test its generalizability. Adding domain adaptation techniques or fine tuning with data specific to each domain could improve its usefulness.
5. **Real-Time Sentiment Analysis:** The project focuses on offline evaluations with a static dataset. An interesting extension could involve developing real time sentiment predictions for streaming the data. It includes live tweets, customer feedbacks or online forums and would allow the businesses and platforms to monitor sentiments instantly.
6. **User-Friendly Interface:** Making a lightweight web or mobile application would make the sentiment analysis system available to non technical users as well. The interface could let users paste or upload a review and get instant sentiment predictions with confidence scores.
7. **Enhanced Robustness through Larger Datasets:** Though the IMDb dataset is large and balanced and includes multilingual reviews or informal social media texts may make the model stronger and more versatile. More diverse data may help manage spelling errors, abbreviations and different cultural contexts in expressing sentiments.

By following the above rules, the project can become a more effective, context aware and user friendly sentiment prediction system that works well in various domains and applications.

## 6.3 Closing Remarks

The ‘Sentiment Prediction of Text Reviews’ project shows what machine learning can do in natural language processing (NLP) and opinion mining. By using proper approach in data pre-processing, feature extraction and model training, this project successfully classifies movie reviews with high accuracy and reliability despite facing challenges with sarcasms, contextual ambiguities and linguistic complexities, the project lays a strong groundwork for further progress in sentimental analysis.

The project shows promises of or automating large scale opinion mining. It also opens up opportunities for useful applications in customer feedbacks management, brand monitoring and real time market sentiment analysis. By developing the project with future improvements, sentiment prediction systems can become more aware of contexts, broader in use and easier to access. It will help individuals, businesses and researchers better understand human opinions and emotions expressed in texts.

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## Glossary

- **Sentiment Analysis:** A natural language processing (NLP) task involves figuring out the emotional tone, like positive or negative, found in a piece of text.
- **IMDb Dataset:** A large publicly available dataset of online movie reviews, containing 50,000 labeled reviews equally split between positive and negative classes.
- **Text Preprocessing:** A series of steps is applied to raw text to prepare it for analysis. This includes removing unwanted characters, tokenization, stop word removal, lemmatization and lowercasing.
- **Tokenization:** The process of splitting text into smaller units (tokens), usually words, which serve as the basic input for feature extraction.
- **Stop Words:** Common words, such as “the,” “is,” and “and,” are often removed from text before feature extraction. They do not add much to the meaning or sentiment.
- **Lemmatization:** The process of reducing words to their base or dictionary form (e.g., running → run), ensuring consistency across similar word forms.
- **Count Vectorizer:** A feature extraction method that represents text documents as vectors based on the frequency of words appearing in them.
- **TF-IDF (Term Frequency – Inverse Document Frequency):** A vectorization method that assigns weight to words based on how often they appear in a document compared to their frequency in all documents. It focuses on unique and meaningful words.
- **Dummy Classifier:** A simple baseline model that makes predictions using basic strategies, such as random guessing or predicting the most frequent class, used for performance comparison.
- **Logistic Regression:** A widely used classification algorithm that models the probability of a binary outcome based on input features.
- **Random Forest Classifier:** An ensemble classification algorithm that builds multiple decision trees and combines their outputs for improved prediction accuracy.
- **Naïve Bayes:** A family of probabilistic classifiers based on Bayes’ theorem that assumes independence between input features, often effective for text classification tasks.

- **Voting Classifier:** An ensemble method that combines predictions from multiple classifiers (e.g., Logistic Regression, Random Forest, Naïve Bayes) by majority vote or weighted probabilities.
- **Precision:** A measure of how many of the instances predicted as positive were actually positive. Defined as  $\text{true positives} / (\text{true positives} + \text{false positives})$ .
- **Recall:** A measure of how many actual positive instances were correctly identified. Defined as  $\text{true positives} / (\text{true positives} + \text{false negatives})$ .
- **F1-Score:** The harmonic mean of precision and recall, used as a balanced metric to evaluate classification performance.
- **Confusion Matrix:** A tabular representation of a classifier's performance showing counts for true positives, true negatives, false positives and false negatives.
- **Overfitting:** A situation where a model learns patterns specific to the training data, including noise, resulting in poor generalization to new, unseen data.
- **Baseline Performance:** A reference performance level of a simple or naive model, used to compare improvements made by more complex models.
- **Sentiment Classes:** The target labels used in this project, i.e., Positive and Negative, representing the polarity of the movie reviews.

## Summary

The “*Sentiment Prediction of Text Reviews*” project aims to develop a machine learning-based system for automatically classifying IMDb movie reviews into positive or negative sentiment categories. The project focuses on binary sentiment analysis using a publicly available IMDb dataset, which contains 50,000 balanced text reviews (25,000 positive and 25,000 negative). The dataset was provided in CSV format and required extensive preprocessing steps, including removal of unwanted characters, tokenization, stop word removal, lemmatization and lowercasing, to prepare the data for effective machine learning.

Feature extraction was done using Count Vectorizer and TF-IDF Vectorizer. This process converted raw text into structured numerical forms suitable for training models. Several machine learning algorithms were tested, including Logistic Regression, Random Forest, Naïve Bayes, a Voting Classifier and a Dummy Classifier for baseline comparison. The tuned Logistic Regression model gave the best performance, reaching an accuracy of 89.3% and an F1-score of 0.91. This result showed that it effectively captured sentiment-bearing patterns from the text, performing better than the other classifiers.

The project highlighted the importance of preprocessing and feature representation in natural language processing tasks. It showed that frequency-based methods (Count and TF-IDF) work well for structured datasets like IMDb reviews. We evaluated performance using multiple metrics: accuracy, precision, recall, F1-score and confusion matrices. These metrics offered deeper insights into performance for each class. Error analysis indicated that misclassifications often resulted from sarcasm, mixed sentiments or specific vocabulary.

Despite the solid results, some challenges persist. Simple vectorization methods cannot fully capture contextual meaning. There are risks of overfitting in high-dimensional feature spaces and handling complex language structures can be difficult. Nonetheless, the project provides a strong base for future improvements through better methods, such as contextual embeddings (Word2Vec, GloVe, BERT) and deep learning architectures (like LSTMs, Transformers).

In conclusion, the “*Sentiment Prediction of Text Reviews*” project demonstrates the potential of machine learning in automating opinion mining from large-scale text reviews. By delivering high performance with traditional ML models and highlighting key challenges, the project opens avenues for future exploration in deep learning, real-time applications, cross-domain adaptability and the development of user-friendly sentiment analysis systems.

## Further Reading

### **Deep Learning for Sentiment Analysis: A Comparative Review**

This paper provides a comprehensive review of deep learning techniques applied to sentiment analysis, comparing RNNs, CNNs and transformer-based approaches.

Author: Y. Zhang, R. Wang

ISSN: 2666-2919 (Online)

Journal: *Information Fusion*

Volume: 66

Published: February 2021

Link: [www.journals.elsevier.com/information-fusion](http://www.journals.elsevier.com/information-fusion)

### **Sentiment Analysis: Mining Opinions, Sentiments and Emotions**

A detailed book that discusses foundational concepts, techniques and applications of sentiment analysis in depth.

Author: Bing Liu

Publisher: Cambridge University Press

Edition: Second

Published: 2020

ISBN: 9781108475325

Link: [www.cambridge.org](http://www.cambridge.org)

### **Advances in IMDb Sentiment Classification Using Transformers**

This article explores the use of state-of-the-art transformer-based architectures such as BERT and RoBERTa on the IMDb dataset, demonstrating improvements over classical ML methods.

Author: A. Vaswani et al.

Journal: *Proceedings of ACL Anthology*

Volume: 12

Published: May 2022

Link: [www.aclanthology.org](http://www.aclanthology.org)

### **Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems**

A practical guidebook covering preprocessing, embeddings and deep learning methods for NLP tasks including sentiment prediction.

Authors: S. Raj, A. Ananthakrishnan, R. Gupta, M. Bhattacharyya

Publisher: O'Reilly Media

Published: 2020

ISBN: 9781492054047

Link: [www.oreilly.com](http://www.oreilly.com)

**Scikit-learn User Guide: Machine Learning in Python**

An official open-source documentation providing detailed insights into machine learning algorithms, preprocessing and evaluation metrics used in projects like sentiment analysis.

Authors: F. Pedregosa et al.

Publisher: INRIA

Maintained: Continuously Updated

Link: [https://scikit-learn.org/stable/user\\_guide.html](https://scikit-learn.org/stable/user_guide.html)

# Plagiarism Report

Kapish Shukla

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Page 2 of 58 - Integrity Overview

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